Chapter 1: Introduction to Python

Python Programming Notes | Beginner Study Chapter

What is Python?

Python is a high-level programming language with readable syntax. It is used for automation, websites, data science, machine learning and artificial intelligence. A Python program is a sequence of instructions that the Python interpreter executes.

Your first program

The print() function displays output. Text is written inside quotes. Python files normally use the .py extension. Run a saved file using python filename.py.

Indentation

Python uses indentation to group statements inside conditions, loops and functions. Use four spaces for each indentation level. Incorrect indentation can cause an IndentationError.

Input and errors

input() reads text from the user and returns a string. Use int() or float() when numerical input is needed. A syntax error means the code structure is invalid. A runtime error occurs while a program is running.

Try this code

```
print("Hello, Python!")
name = input("Enter your name: ")
print("Welcome", name)
```